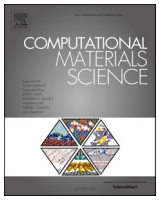
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Full Length Article

Enlightenment the dynamic behavior of norbornene–modified 'click' 4–arm polyethylene glycol hydrogel: Delving into framework properties and transport properties through molecular dynamics simulations

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ABSTRACT

The thiol-norbornene cross-linked 4-arm Polyethylene Glycol (PEG) 'click' hydrogel is a synthetic, sustainable, and bio-inspired polymer extensively used in biomedical applications. Experimental techniques are widely used to study this system, whereas all-atom molecular dynamics simulations, which offer insight into the dynamic behaviours of the system, are never used to study this system. Three models of thiol-norbornene cross-linked PEG 'click' hydrogel and a base PEG hydrogel are crafted. It is immersed in water to study its structural and transport properties. Structural properties are analysed through root-mean-square deviation (RMSD), radial distribution function (RDF), hydrogen bonds (H-bond), and stress–strain behavior. The RMSD reveals that the norbornene component enhances hydrogel stability compared to the base model. The RDF illustrates interactions between oxygen in PEG chains and water. H-bond results underscore PEG's strong H-bond acceptance. The norbornene-functionalized crosslinked PEG hydrogel displays maximum H-bonding. It demonstrates a superior swelling ratio attributed to freezing and non-freezing water effects, indicating high stability and potential suitability for applications. Mean square displacements (MSD) unveil the diffusion coefficients of the hydrogel. The base hydrogel model shows the diffusive behavior, and the diffusion coefficient is 1.97×10^{-10} m^2/s. The base model has the highest MSD value compared to other systems. The thiol-norbornene crosslinked click hydrogel has the highest Young's modulus, which signifies the stiffness of the material. Findings illuminate thiol-norbornene 'click' PEG hydrogel behaviour, emphasising chemical cross-linking's crucial role in advanced biomedical applications.

1. Introduction

Polyethylene glycol (PEG) hydrogels are sustainable and biodegradable hydrogel that is composed of cross–linked PEG chains[1]. These hydrogels have gained popularity due to their low toxicity, solubility, biocompatibility, and versatility, making them suitable for various biomedical applications, such as tissue engineering[2], drug delivery[3], and wound healing [4]. The single–arm PEG demonstrates a lower cross–linking efficiency, leading to a network structure that is less robust and stable[5]. This leads to decreased mechanical strength, lower gel stability, and increased susceptibility to degradation. Furthermore, the single–arm PEG chain has a lower water content, limiting the diffusion of nutrients and bioactive molecules[6]. To overcome these challenges, multi–arm PEG chains are required. The 2–arm PEG[7], 4–arm PEG[7–9], 6–arm PEG[9], and 8–arm PEG[10,11] chains are synthesized. Among them, the 4–arm PEG chain has gained more interest due to its enhanced functionality and increased stability compared to the other chains[7–9]. The cross–linking mechanism is used to further modify the 4–arm PEG hydrogel properties for a desirable application[12,13]. The choice of cross–linking is a crucial step in the modification of the 4–arm PEG hydrogels because it determines the physical and chemical properties of the hydrogel, such as degradation rate, mechanical strength, and release kinetics[13]. Several different crosslinking methods are used to modify the 4–arm PEG hydrogels. Each method has its own advantages and disadvantages. There are three methods of crosslinking the 4–arm PEG hydrogels: physical crosslinking [14], enzymatic crosslinking[15], and chemical crosslinking[8]. Physical crosslinking, using methods such as freeze–thawing[14]. However, this method can be time–consuming, and it requires specialized types of equipment[16]. To enhance the control crosslinking, enzymatic crosslinking can be used. It uses enzymes to catalyze the formation of covalent bonds between the 4–arm PEG chains[15]. This method has several

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advantages, such as being environmentally friendly, biocompatibility, and controlled crosslinking, but this method has several disadvantages, also like it requires specific enzymes and it is more complicated than other methods[15,17]. On the other hand, chemical crosslinking is a fast and simple method of modifying the 4–arm PEG hydrogels, and it allows for the modification of the hydrogel properties through the choice of crosslinking agent. There are several chemical crosslinking agents available, such as glutaraldehyde[18], ethylene glycol diglycidyl ether [19], thiol–ene agent[7–9], etc. The thiol-ene crosslinking agent has increasingly captured attention in recent years due to the versatility and adaptability of the reaction [7–9]. The thiol–ene click chemistry is a type of reaction between a thiol (sulfur–containing) molecule and an alkene (carbon–carbon double bond) molecule. The reaction is fast, efficient, and highly selective, and it is widely used for the formation of covalent bonds in various materials, including the 4–arm PEG hydrogels [20]. The 4–arm PEG hydrogels made through thiol–ene click chemistry are also biocompatible and non–toxic, making them suitable for use in biomedical applications. The thiol–ene crosslinked 4–arm PEG hydrogel provides several benefits over traditional hydrogel fabrication methods, including improved mechanical properties, increased stability, facile modification with various functional groups, and degradation rate. These properties can be achieved by adjusting the molecular weight, crosslinking density, and chemical structure of the 4–arm PEG chains [7–9]. Thus, the Norbornene is used to functionalize the 4–arm PEG, forming a hydrogel through thiol–ene click chemistry.

The main goal is to produce a durable, stable crosslink network that is highly reactive, stable, and biocompatible for biomedical uses. Thus, the PEG undergoes crosslinking with norbornene using a radical–mediated thiol–norbornene reaction triggered by light. They find that the degradation of norbornene–thiol crosslink PEG hydrogels can be regulated by modifying cross–linkers with different hydrolysis susceptibility and incorporating steric hindrance, hydrophobicity, and electron–withdrawing moieties. Additionally, the degradation rates can be adjusted by manipulating macromer compositions[8]. The thiol–ene 'click' 4–arm PEG hydrogels possess desired mechanical attributes similar to soft tissues, maintaining stability during cell culture while allowing viable encapsulated cells. The gels exhibit spatial control over matrix properties through selective patterning of biochemical cues while ensuring cell viability[21]. Although the experimental techniques characterize the properties of 4–arm PEG hydrogel[7–9], the precise measurement, and reproducibility are hindered by the hydrophilic nature of the hydrogel's network. Experimental methods cannot elucidate the structural orientations in the aqueous medium, non–bonded interactions like hydrogen bonding (H-bonding), and transport properties. H-bonding is crucial in hydrogel applications and can be accurately studied through theoretical investigation. Molecular dynamics (MD) simulations replicate gel microstructures and provide molecular–level insights into material behavior, including H-bonding, diffusivity, conductivity, and stress–strain behavior of PEG hydrogel.

MD simulations have explored the interaction between dithymoquinone (DTQ) and PEG to assess its drug delivery potential. Results showed that PEG 60 is the most suitable polymer for encapsulating DTQ, enhancing drug solubility and reducing flexibility in the presence of water [22]. Additionally, PEG hydrogels, reinforced with cellulose nanofibrils, are analyzed for mechanical stability using MD simulations. The simulations demonstrated that incorporating 3 % cellulose significantly improved the ultimate strength and Young's modulus of the hydrogel, making it suitable for wound healing applications [23]. In another case, MD is employed to study the rheological properties of poly (ethylene glycol) and poly(propylene glycol) hydrogels in enhanced oil recovery (EOR) applications, confirming their suitability for reducing interfacial tension and improving oil displacement efficiency [24]. Mixed matrix membranes (MMMs), incorporating silica nanoparticles, have also been investigated via MD simulations. The addition of silica enhanced membrane permeability and selectivity for gases like CO_2 and CH_4, supporting the development of more efficient gas separation systems [25]. Furthermore, MD simulations have provided insights into hydrogel network structures, such as mesh sizes and pore distributions, which are critical for understanding swelling behavior and solute transport [26]. MD simulations have been employed to investigate interactions between alginate gel and PEG-grafted polyamide membranes to understand antifouling properties. Results demonstrated that PEG coatings retain a hydration water layer, which creates strong repulsive hydration forces, preventing alginate adhesion to the membrane surface [27]. MD studies have analyzed the degradation of PEG-Terpolymer toluene diisocyanate based polymer systems in energetic applications, revealing that increased degradation leads to enhanced compatibility with energetic plasticizers and improved mechanical properties [28]. Similarly, methacrylic acid porous hydrogels have been simulated for their thermal and mechanical properties, demonstrating that increasing the volume fraction of the polymer enhances hydrogel stability [29]. MD is used to investigate the behavior of various hydrogel systems, including those based on PEG and poly(N-isopropylacrylamide) (PNIPAAm). These simulations provide insights into the effects of parameters such as polymer concentration, molar mass, and temperature on the hydrogel network structure, water-polymer interactions, and drug encapsulation and release [30]. By combining MD with experimental techniques, scientists can gain a deeper understanding of the structure–property relationships in hydrogels, enabling the rational design of these materials for biomedical applications like drug delivery and tissue engineering [31]. In the field of bone cement, MD simulations have been utilized to study the effects of adding magnesium nanoparticles to calcium phosphate cements, demonstrating enhancements in mechanical and thermal behavior [32]. Researchers employ all-atom MD (AAMD) simulations to examine the phenomenon of "negative energetic elasticity" in these materials, which refers to a negative contribution of the energy to the overall elastic modulus. The simulations involve modeling a PEG network in water, evaluating the temperature-dependence of shear stress under constant-volume conditions, and using statistical approaches to estimate the parameter T_E, which is an indicator of the negative energetic elasticity. By comparing the results under constant-pressure and constant-volume conditions, the researchers validate their simulation methodology and provide insights into the complex mechanical properties of PEG hydrogels at the atomic scale [33]. Looking into the MD simulation literature, the PEG hydrogel simulations provide valuable atomic-level details about structure, bonding, and material behavior, guiding experimental efforts and accelerating the development of improved hydrogel formulations for various applications. However, compared to the linear PEG structure, the 4-arm-PEG has more importance, yet it has not been studied using MD simulation.

The study addresses a significant knowledge gap by utilizing MD simulation to investigate the dynamic behavior of 4-arm-PEG and its thiol-norbornene chemically crosslinked polymer matrix in the aqueous media. The primary objective is to construct an atomistic model of the thiol-ene functionalized 4-arm PEG gel, incorporating appropriate water content. This approach allows for a comprehensive examination of both structural and transport properties of the hydrogel system. The simulation results are analyzed using various computational techniques. Structural properties are evaluated through root mean square deviation calculations, analysis of the H-bond network, radial distribution function, and stress–strain analysis. These methods provide insights into the gel's conformational stability, intermolecular interactions, spatial distribution of atoms, and mechanical behavior. Transport properties are assessed by calculating mean square displacements, which offer information on the mobility of different components within the hydrogel.

2. Computation methodology

The atomic structure of ethylene glycol monomer is first drawn using the Avogadro package [34]. Thereafter, the ethylene glycol monomer is replicated 20 times to form a polymeric chain through in–house programming which is suggested in the literature [35–37]. Thereafter, the

last hydroxyl moiety in the PEG chain is modified with norbornene to enhance the property of the polymeric materials [7–9]. The thiol moieties are introduced to the "ene" functional norbornene group to produce PEG functionalized with thiol–ene. Fig. 1 displays the typical structures for each of these three polymer chain structures. Later the four PEG chains are attached to the carbon atom in the Avogadro package to prepare the final structure of the 4–arm PEG (Figure S1(a) of the supplementary materials (SM)). Similarly, the norbornene functionalized PEG (NPEG, Figure S1(b) of the SM) and dithiothreitol–norbornene functionalized PEG (TPEG, Figure S1(c) of the SM) are prepared. Therefore, each chain contains four arms, each arm containing 20 repeating units of ethylene glycol monomer with norbornene and dithiothreitol–norbornene functionalization at the end of each chain (Figure S1 of SM). The dithiothreitol serves as a crosslinker which is confirmed by experimental studies. As a result, the NPEG and TPEG alternate chains are linked together to create a thiol–ene crosslinked PEG gel (CPEG, as depicted). Several of these crosslinked gels are connected to form a network, which is shown in Fig. 1(d).

The objective of this study is to examine the dynamic behavior of PEG and its thiol–ene functionalized polymers. To achieve this, the Packmol package is used to insert the different configurations of the polymer chains which are namely PEG, NPEG, TPEG, and CPEG, are inserted into the aqueous medium[38]. The TIP4P water model is utilized for the solvent molecules in the hydrogel system[39]. The PEG, NPEG, TPEG, and CPEG systems have a total number of atoms present: 46886, 40562, 39782, and 89774, respectively. The number of polymer atoms, the number of water molecules in each system and the dimensions of the box are provided in Table S1 of the SM. The gasteiger method is used to generate partial charges on each atom[40]. The interactions between bonded and nonbonded atoms are computed using the Amber14SB force field[41]. This force field is an updated version of the AMBER99SB force field, which is widely used in macromolecule simulation [42–45]. The functional form of the Amber14SB[41] force field is represented by equation (1). The non-bonded and bonded force field parameters of polymer and water are taken from the Amber14SB forcefield. However, the non-bonded and bond equilibrium length and bond constant is given in Table S2-3 of the SM. The angle, proper dihedral and improper dihedral parameters are taken from the Amber 14SB force field literature [40]. The chronological trajectories are generated by MD simulations. The GROMACS–2022.3 open–source package is used to perform the MD simulation [46].

$$V_{total} = \sum_{bonds} k_b (r - r_0)^2 + \sum_{angles} k_\theta (\theta - \theta_0)^2 + \sum_{dihedrals} k_\chi \left[1 + \cos(n\chi - \delta_\chi)\right] + \sum_{improper} k_\Psi [1 + \cos(n\Psi - \delta_\Psi)\,] + \sum_{i=1}^{N-1} \sum_{i<j}^{N} \left\{ 4\varepsilon_{ij} \left[(\frac{\sigma_{ij}}{r_{ij}})^{12} - (\frac{\sigma_{ij}}{r_{ij}})^{6} \right] + \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} \right\} \quad (1)$$

where, V_{total} encompasses different energy components, including bond stretching, angle bending, torsional (proper dihedral and improper dihedral), van der Waals interactions, and columbic interactions. The Amber14SB force field defines parameters such as bond stretching constant (k_b), bond length (r), standard bond length (r_0), bending angle energy constant (k_θ), angle between molecules (θ), standard angle (θ_0), proper dihedral constant (k_χ), dihedral angle between planes (χ), multiplicity (n), phase shift (δ_χ), improper dihedral constant (k_ψ), improper dihedral angle between planes (ψ), multiplicity factor (n), phase shift (δ_ψ), depth of potential wall (ε_{ij}), distance at which potential is zero (σ_{ij}), repulsive term at short distance (r_{ij}^{-12}), attractive term at long distance (r_{ij}^{-6}), permittivity of vacuum (ε_0), charge of the particles (q_i and q_j), and distance between charge particles (r).

The system dynamics are acquired by the sequential methodologies of energy minimization, equilibration, and production. The energy minimization method is used to adjust the atom's position in the system in such a way which effectively removes the high energy contacts and enables the convergence to a more physically realistic energy minimum system in the MD simulations. The steepest descent algorithm is used for energy minimization. Its simplicity and effectiveness in quickly reducing large forces. It works by iteratively moving the system along the direction of the steepest negative gradient of the potential energy, allowing rapid convergence toward a local minimum widely used in the literature [47,48]. The energy cut–off value for the minimization is 10 KJ/mol·nm. During the minimization process, a step size of 0.001 is used with a maximum of half a million steps. The periodic boundary conditions are utilized in all dimensions. A cut–off distance of 10 Å is used for the computation of van der Waals dispersion interactions. A cut–off distance of 12 Å is employed to compute the short–range electrostatic interactions. The long–range electrostatic interactions are commutated using the particle mesh Ewald technique [49]. Directly summing the electro-static interactions is computationally expensive because they decay slowly and require considering distant particles. PME addresses this by splitting the electrostatic potential into short-range and long-range components. The short-range interactions are computed in real space, while the long-range part is handled in reciprocal space using a fast Fourier transform on a mesh grid. This method significantly reduces the computational cost, allowing accurate and efficient calculation of electrostatic forces in periodic systems without truncation errors, and is widely used in the literature [50,51]. During the equilibration and production run, the leapfrog algorithm is employed to integrate Newton's second law of motion with a time step of 2 fs. The leapfrog algorithm offers stability and accuracy in numerical simulations. A 2-fs time step is used in all-atom MD simulations because it balances accuracy with computational efficiency, as suggested in the literature [35,36,52,53]. This time step is sufficient to capture rapid oscillations of hydrogen atoms without causing instability, as constraints are applied to these bonds. Using a 2-fs time step allows for faster simulations, effectively doubling the speed compared to a 1-fs time step while still maintaining accurate results. Additionally, it adheres to Nyquist's theorem, ensuring that the simulation captures all relevant molecular motions without aliasing [54]. This approach provides a good compromise between detailed accuracy and computational cost. The velocity rescale [55] thermostat is used to constant the system temperature at 310 K throughout the dynamics, which are widely studied and thoroughly applicable to macromolecule systems simulations[56–58]. This thermostat controls by adjusting the velocities of particles. It works by periodically scaling the velocities of all particles to ensure that the system's temperature matches the desired value. This method efficiently corrects deviations from the target temperature while minimizing disturbances to the system's dynamics. By applying a time-dependent scaling factor, the velocity rescale thermostat allows the system to reach and stabilize at the target temperature quickly, helping to maintain thermal equilibrium throughout the simulation [55]. During the equilibration process, the linear constraint solver (LINCS) algorithm is used to suspend the motion of the hydrogen moieties [59]. The LINCS algorithm efficiently maintains constraints on H-bonds with other moieties, which reduces the computational cost and faster the equilibration process [53,60]. The equilibration process consists of a five ns simulation in the NVT ensemble, succeeded by a 125 ns simulation in the NPT ensemble. During the NVT equilibration, the initial velocity is generated using the Maxwell–Boltzmann distribution, where the velocity is assigned by the reference temperature and mass of the system[61]. During the NPT equilibration phase, the pressure is kept constant at 1.013 bar. The Parrinello–Rahman method is used to keep the constant pressure throughout the dynamics of the system by changing the simulation box volume[62]. The Parrinello–Rahman barostat dynamically adjusts the simulation cell's shape and volume to ensure pressure stability. This method enables accurate representation of isotropic pressure conditions and is particularly effective for simulations involving complex or varying system geometries. By adapting the cell dimensions, it provides a more realistic and flexible approach to pressure control [62]. During NPT equilibration, the temperature and density data are given in

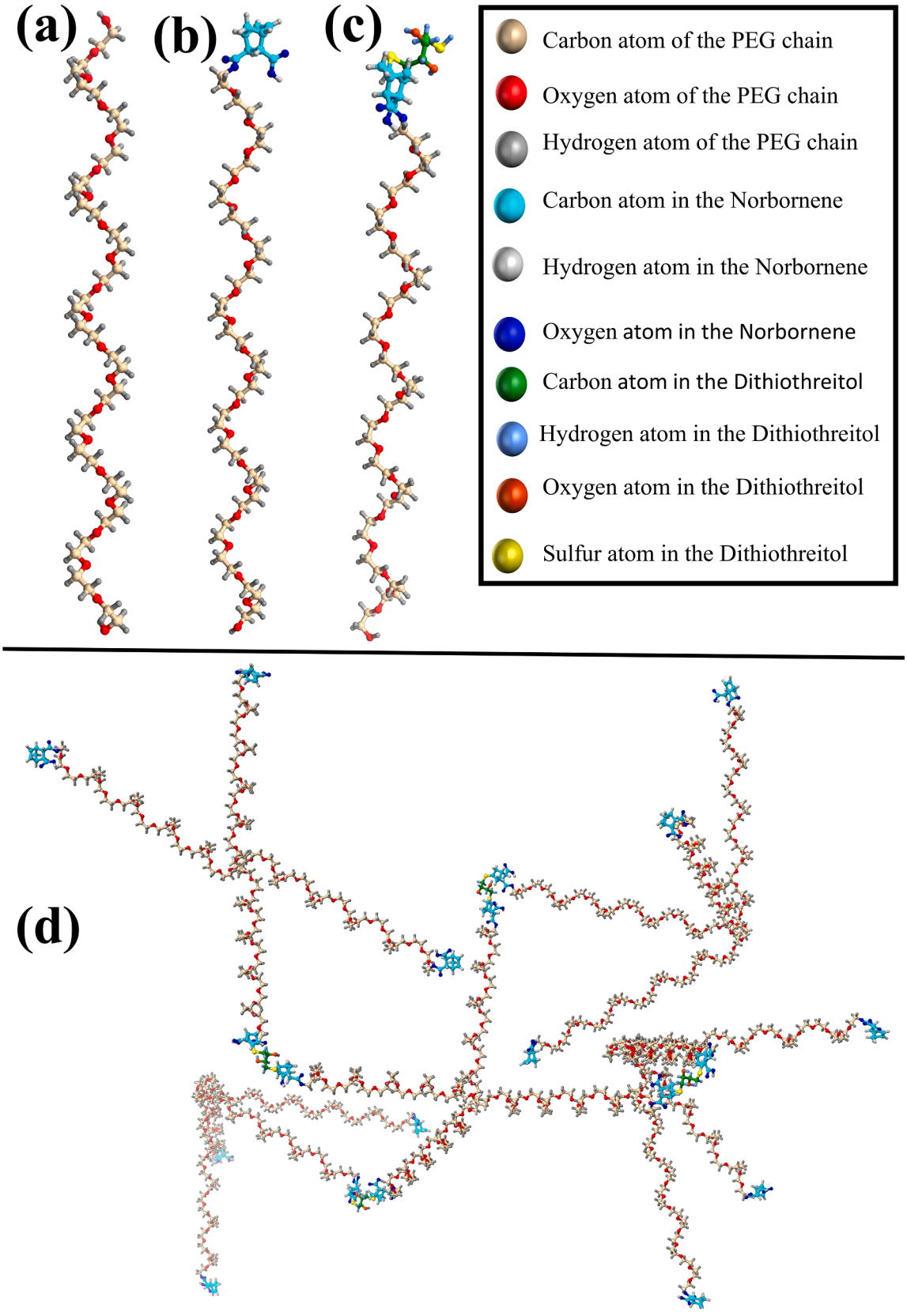


Fig. 1. The initial structure of polymer (a) 4–arm PEG, (b) 4–arm NPEG, (c) 4–arm TPEG, and (d) 4–arm CPEG.

the Figure S2 of SM. Figure S2 suggests that the temperature and density of the system fluctuate around the mean value, indicating that the system is well equilibrated. After the system's dynamic equilibrium is achieved, the isobaric isothermal ensemble is utilized for the production run. The production run process consists of 50 ns of MD simulation. The non-equilibrium MD simulation is performed using the $N\sigma_x\sigma_y$ ensemble. During this ensemble, the deformation rate is set to 0.2 nm/ns in the z-direction for the 15 ns. During this ensemble, the pressure couple type is set to the anisotropic. This allows for independent scaling of the simulation box dimensions in all directions. This is essential when simulating stress–strain behaviour, as the system may experience different deformations along different axes. The chronological trajectory is saved at a ten ps time interval, which is further used for the analysis. This assessment confirms that the establishes approach is sufficient for generating accurate results through statistical averaging of a solitary trajectory for the polymer chains in the liquid phase. The detailed NPT equilibration and non-equilibration parameters are given in SM. The visualization of the chronological trajectory is done by the VMD[63] and the Chimera X open–source package [64,65]. The chronological snapshot of the PEG, NPEG, TPEG and CPEG system are given in Figure S3, Figure S4, Figure S5, and Figure S6 of supporting documents, respectively. The Final simulation snapshot is given in Fig. 2. The analysis of the trajectory is done by the in–house Python scripts.

2.1. Analysis of trajectory

2.1.1. Root mean square deviation (RMSD)

RMSD plays a crucial role in MD simulations, demonstrating its significance in assessing the accuracy and reliability of simulated structures. It quantifies the deviation between the positions of atoms in the simulated system and a reference structure. The RMSD provides insights into the structural stability, conformational changes, and overall quality of the simulated trajectory. Structural stability in a hydrogel is essential because it directly affects the material's mechanical strength, durability, and performance in different applications A stable structure ensures that the hydrogel can maintain its shape and integrity under stress or environmental changes, such as temperature or pH fluctuations. This stability allows the hydrogel to retain its ability to absorb and hold water, maintain its network structure. The common formula used to determine the RMSD of specific atoms in a molecule from a reference structure is

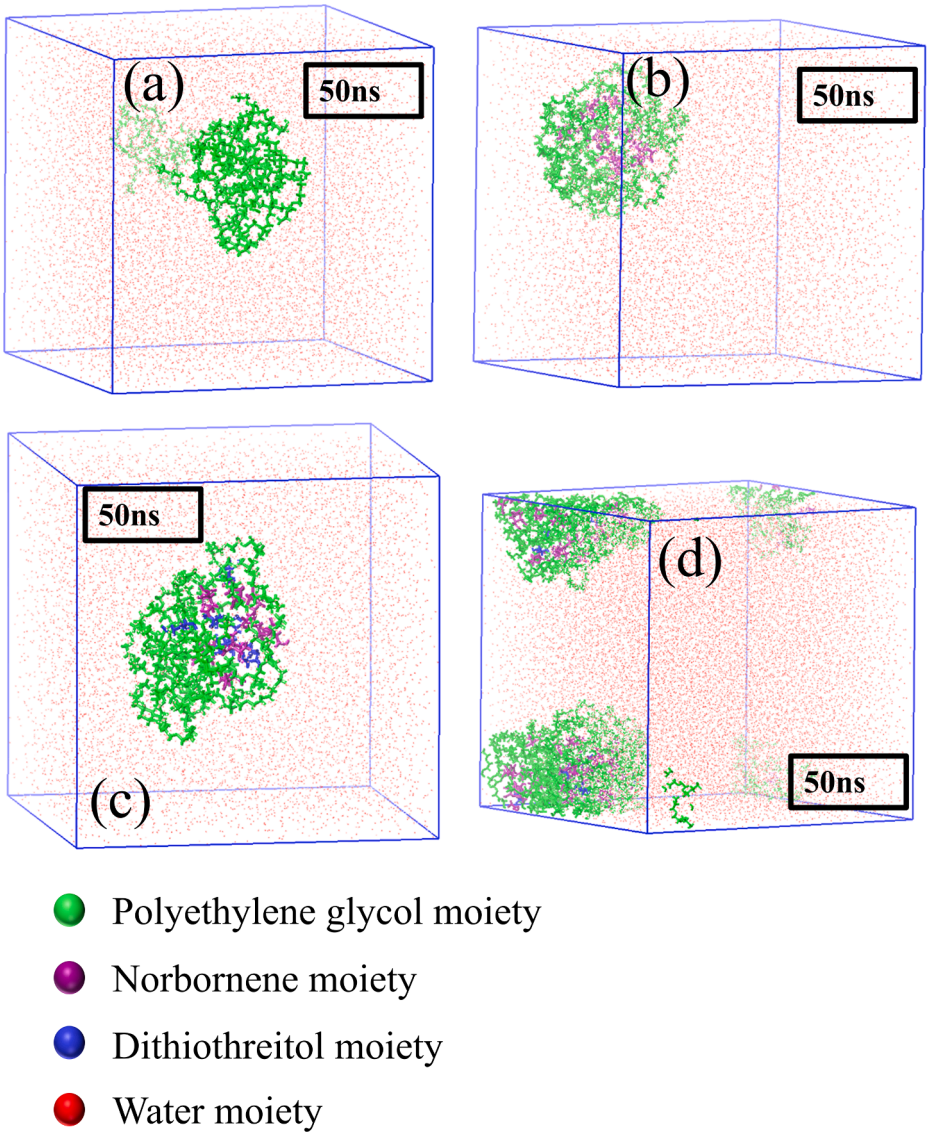


Fig. 2. The final structure of hydrogels at the end of 20 ns (a) 4–arm PEG hydrogel, (b) 4–arm NPEG hydrogel, (c) 4–arm TPEG hydrogel, and (d) 4–arm CPEG hydrogel.

$$RMSD(t_1,t_2) = \left[\frac{1}{M}\sum_{i=1}^{N} m_i \|r_i(t_1) - r_i(t_2)\|^2\right]^{\frac{1}{2}} \quad (2)$$

$$M = \sum_{i=1}^{N} m_i \quad (3)$$

Where the ith particle's mass is given by m_i, N is the total number of particles present in the system, and r_i is the position at the time t_1 and time t_2.

2.1.2. Radial distribution function (RDF)

The calculation of the RDF holds significant importance in MD simulations. RDF analysis allows for a comprehensive understanding of the spatial arrangement and interactions between particles in a system. By quantifying the probability of finding particles at different distances from a reference particle. RDF provides insight into how molecules or atoms are distributed relative to one another, revealing information about local structure and solvation effects. The mathematical formulation for the RDF is as follows:

$$g(r) = \frac{1}{N}\left(\frac{V}{4\pi r^2 dr}\right)\sum_{i=1}^{N}\sum_{i\neq j}^{N}\delta(|r_i - r_j| - r) \quad (4)$$

Where $g(r)$ is the RDF, N is the total number of particles in the system. V is the total volume of the simulations box, r is the distance between the reference particle and the other particle in the system, dr is the width of the radial bin, $\delta(z)$ is the Dirac delta function, which equals one if $z = 0$ otherwise, and $|r_i - r_j|$ is the distance between particle between i and j. The RDF calculation involves dividing the simulation box into a series of concentric spherical shells with widths of dr. The particles in the system are then counted within each shell, and the result is normalized by the density of the system to obtain $g(r)$.

The coordination number holds great significance in MD simulations as it provides valuable information about the local environment and interactions of particles within a system. The coordination number represents the number of neighboring particles surrounding a central particle. By calculating the coordination number, one can gain insights into the structure, stability, and bonding characteristics of materials at the atomic or molecular level. It helps in identifying changes in coordination during phase transitions, analyzing the arrangement of atoms in crystals or liquids, and understanding the formation and dissolution of chemical bonds. Furthermore, the coordination number plays a crucial role in studying diffusion, conductivity, and other transport properties in various materials. The coordination number is calculated as

$$CN(r) = 4\pi\rho N\int_0^r g(r)r^2 dr \quad (5)$$

Where CN(r) is the coordination number, ρ is the number of density particles, N is the total number of particles in the system, and $g(r)$ is the radial distribution function. The integral over the spherical shell of radius r counts the number of particles within that distance from the reference particle.

2.1.3. H-bonding

H-bonding plays a pivotal role in hydrogels, showcasing their utmost importance in their structural and functional properties. These unique interactions occur between the electronegative moieties present in the polymer chains and the hydrogen atoms of water molecules. The strength of H-bonds allows hydrogels to maintain their three–dimensional network structure and the degree of crosslinking, facilitating their high–water content and excellent swelling properties. Additionally, H-bonding enhances the stability and mechanical strength of hydrogels. The ability of H-bonding to control the diffusion of molecules and ions within hydrogels further accentuates their significance in achieving desired functionalities. The geometrical criteria are used to calculate the formation of a H-bond between an acceptor atom, A, and a donor atom, D. The H-bond is considered to have formed when the hydrogen donor to acceptor angle (α_{HDA}) is less than 30°. Additionally, the distance between the donor and acceptor atoms (r_{DA}) should be less than 0.35 nm. The dynamics of H-bonding are calculated by utilizing the definition of the intermittent H-bond autocorrelation function as

$$C_{HB} = \frac{\langle H(\tau)H(\tau + t)\rangle}{\langle H(\tau)H(\tau)\rangle} \quad (6)$$

Where C_{HB} is the number of H-bonds, $H(t)$ is assumed to be either 1 or 0, depending on whether the chosen pair forms a H-bond at that specific moment.

2.1.4. Swelling ratio

The swelling ratio is a critical characteristic of hydrogels as it reflects the material's ability to absorb water and expand, directly influencing its mechanical properties, permeability, and functionality. A high swelling ratio indicates greater water uptake, which enhances the hydrogel's flexibility and softness, making it suitable for biomedical applications like tissue engineering or drug delivery. Polymer water absorption occurs through two mechanisms: the incorporation of freezing water and the uptake of non–freezing water. Non–freezing water refers to water molecules directly bonded to the polymer chain via H-bond while freezing water represents the relatively weak interaction between water and polar functional groups [66]. The number of water molecules connected to non–freezing water through H-bonds defines freezing water. Both types of water are utilized in computing the swelling ratio. The swelling characteristic can be calculated as follows:

$$SR(\%) = \frac{W_{eq} - W_i}{W_i} \quad (7)$$

Where W_{eq} is the weight of hydrogel composites after swelling in water, and W_i is the weight of dried hydrogel composites.

2.1.5. Mean square displacement (MSD)

The MSD is a crucial parameter in MD simulations, playing a vital role in characterizing the dynamics of particles or molecules. MSD provides insight into the dynamics of processes, such as self-diffusion. It is calculated by taking the difference between the position of the particle at a given time and its initial position, squaring that value, and then taking the average over all particles in the system. The following equation can be used to determine the MSD.

$$MSD(t) = \left\langle (r_i(t) - r_i(0))^2 \right\rangle = \frac{1}{N}\sum_{i=1}^{N}\left[(r_i(t) - r_i(0))^2\right] \quad (8)$$

Where $r_i(t)$ represents the positions of atom i^{th} at time t, $r_i(0)$ is its *initial* position, and the angled brackets ($\langle\,\rangle$) denote an ensemble average over all particles in the system. Using the Einstein relationship, the slope of MSD can be used to compute the diffusion coefficient D.

$$D = \frac{1}{6}\lim_{t\to\infty}\frac{d(MSD(t))}{dt} = \frac{1}{6N}\lim_{t\to\infty}\frac{d}{dt}\sum_{i=1}^{N}\left[((r_i(t) - r_i(0))^2\right] \quad (9)$$

2.1.6. Young's modulus

Young's modulus is the ratio of stress to strain in a material's elastic region, which is crucial for hydrogel systems. It characterises mechanical properties, indicating a hydrogel's stiffness and ability to maintain structural integrity under load. The mathematical formulae given bellow

$$E = \frac{\sigma}{\varepsilon} = \frac{\frac{F}{A}}{\frac{\Delta l}{l_0}} = \frac{Fl_0}{A\Delta l} \quad (10)$$

3. Result and discussion

3.1. RMSD

The RMSD plot of PEG, NPEG, TPEG, and CPEG polymer backbones are shown in Fig. 3. The RMSD values initially increase progressively over time until it reaches an equilibrium plateau which is observed for all polymers. After the equilibrium plateau reaches, the curve shows minimal fluctuation throughout the dynamics. This behavior is shown because of the hydrogen moieties present in the polymer backbone. The mobility of hydrogen moieties becomes constrained during the equilibration. Upon reaching dynamic equilibrium, the restricted motion of hydrogen moieties is eliminated, resulting in an initial increase in the RMSD value. The RMSD value reveals that the PEG chains exhibit the highest value compared to other polymer chains. Initially, the RMSD value of the PEG chain increases until 16 ns, followed by a period of stable behavior. At the end of the dynamics, the RMSD value for the PEG polymer chain reaches 1.25 nm. This higher RMSD value is attributed to the relatively lower stability of PEG chains in aqueous media than other polymer structures. The stabilization of the chain backbone in aqueous media requires more time than other polymer chains. The NPEG chain exhibits the third–highest RMSD value, which is 1.02 nm at the end of the dynamics. The RMSD value of the NPEG chain increases for 15 ns and then the increases gradually till end of the dynamics. The norbornene is a water insoluble component which reduce the NPEG movement in the aqueous media. This restriction of motion gives NPEG structure more stable. The TPEG shows second highes RMSD value which is 1.16 nm till end of the dynamics. The RMSD increases 10 ns then it fluctuates its mean value. The dithitheritol cross linker enhance the initial stability hydrogel but the final stability reduces. The dithiothreitol is water soluble which enhance the solubility of TPEG structure. As a result, it reduces the stability of hydrogel structure. The CPEG structure shows lowest RMSD which is 0.8 nm till end of the dynamics. Due to the bulkiness of the structure, enhances the stability of the network. However, the RMSD of CPEG slope gradually increasing till end of the dynamics. shows minimal fluctuations until the end of the simulation. Upon functionalization of norbornene into the polymer chain, the RMSD value is reduced by half. This observation suggests that the rigid structure of norbornene exhibits a stabilizing influence on the polymer chains. It effectively mitigates their tendency to collapse or undergo conformational changes throughout the ongoing simulations. The RMSD value of the TPEG chain is lower than the PEG and the NPEG chain. The RMSD value of the TPEG chain in the aqueous media is 1.04 nm at the end of dynamics. The RMSD value progressively increases by three ns over the time periods, followed by insignificant fluctuation towards the end of the dynamics. The dithiothreitol functionalization also stabilizes the polymer backbone, and it requires less dynamic equilibrium time than the PEG polymer. The CPEG polymer shows the lowest RMSD value among other polymer structures. The RMSD value of the CPEG polymer is 0.80 at the end of the simulation. However, the RMSD increases gradually till end of the dynamics. It indicates that the system is undergoing structural changes over time. A steady increase suggests that the hydrogel is slowly deviating from its initial conformation, potentially due to molecular rearrangements or unfolding.

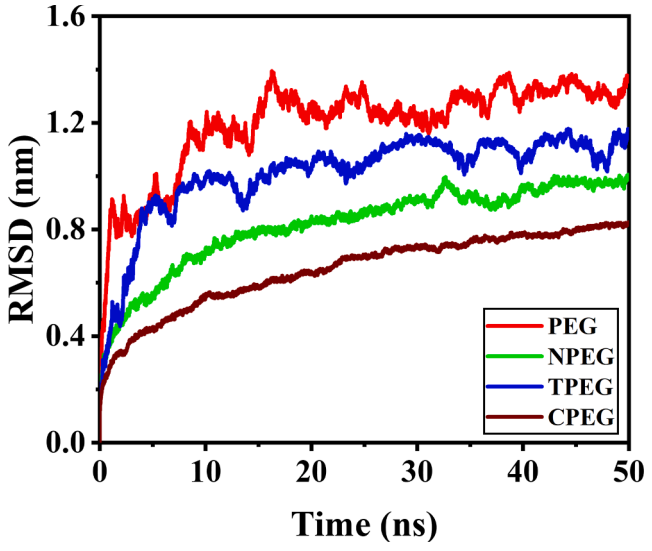


Fig. 3. Polymer chain RMSD to its relative backbone.

3.2. H-bond

H–bonding currently plays a significant role in the structure and properties of hydrogels. The number of H-bonds between polymer to polymer and polymer to solvents is crucial because it helps to maintain the hydrogel 3D structure in the aqueous medium. The hydrogel creates interchain (HB_{inter}), intrachain (HB_{intra}), and polymer–water (HB_{P-S}) H-bonds in an aqueous medium. The presence of neighboring repeat units (HB_{adj}) further categorizes the HB_{inter}. Fig. 4(a) currently depicts the H-bonds of HB_{intra} and HB_{inter} types. The hydroxyl moieties of the hydrogel backbone and the functional moieties of the hydrogel act as H–bond donors and acceptors. The PEG monomer contains a hydroxyl group in

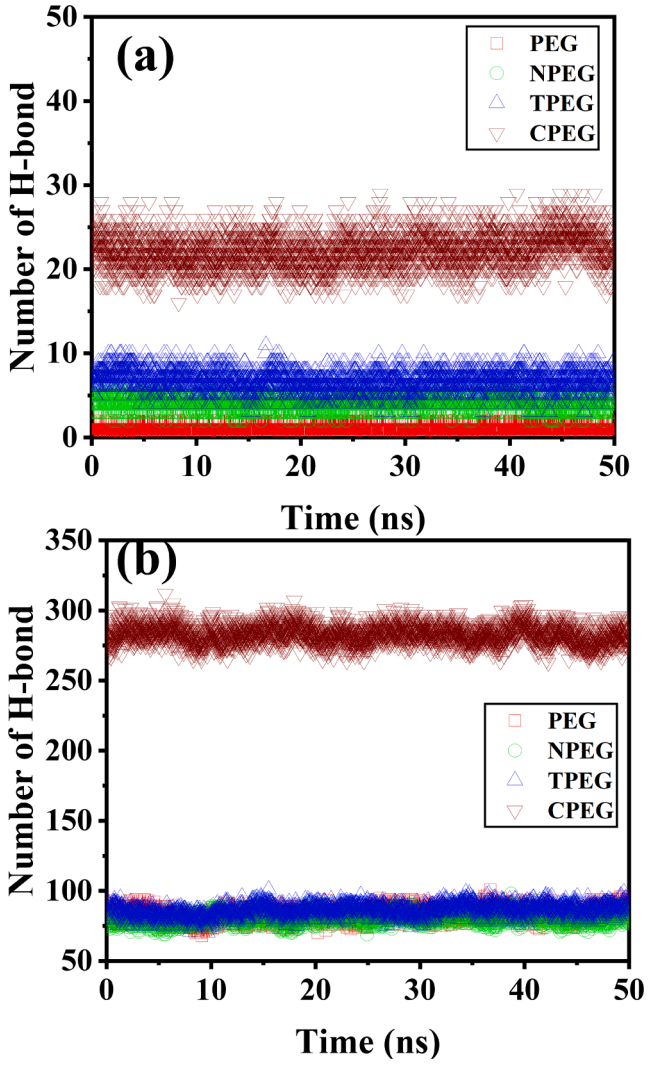


Fig. 4. Number of H-bonds in the system (a) number of H-bonds between polymer to polymer, and (b) number of H-bonds between polymer to solvents.

both terminals, but when it polymerises, the PEG monomer is joined by an ether, which acts as an H-bond acceptor. However, the 4-arm-PEG chains terminate with a hydroxyl group, which acts as an H-bond donor between the polymer chains. During the dynamic property calculation, the slope of all four polymer H–bond curves exhibit a stable throughout the time. This suggests the formation of stable networks throughout dynamics. These findings are suggested by the RMSD results. The order of the H-bonds is PEG < NPEG < TPEG < CPEG. The average number of H–bond between polymer to polymer is 1, 4, 6, and 22, respectively, given in Table 1. It is observed that the 4–arm PEG chain is not an excellent H–bond donor, but it is a suitable H-bond acceptor. This characteristic makes PEG hydrophilic and highly soluble in water, but unlikely to engage in significant self-interactions through H-bonding. The DSC experiments confirm reduces the crystallinity of PEG and polyvinyl alcohol (PVA) blend membrane compare to the PVA and poly (etethylenemine) blend membrane the which signifies the weaker H-bonding interaction. The H-bonding capability of PEG determine the size of the biphasic interphase of water and ionic liquids[67]. The H-bonding acceptor criteria of PEG is helpful for extracting the protein from the pumpkin seeds[68]. The H-bonding of PEG polymer also helpful to extract the riboflavin [69]. However, the PEG is functionalised with norbornene and dithiotheritol. It has a couple of H-bond donors and acceptors, which enhance the number of H-bonds within the polymer. This enhanced number of H-bonds increases the stability of the network in aqueous media.

The H–bond between polymer and solvents is represented by Fig. 4 (b). Fig. 4(b) shows the stable number of H-bonds throughout the dynamics. This suggests the hydrogel network holding the structural intregrity. The order of the number of H-bonds is CPEG ≤ PEG < NPEG < TPEG. The average number of H-bonds between polymer and solvents are 284, 81, 83 and 85, respectively, given in Table 1. The TPEG form highest number of H-bonds with water. The dithiothreitol moiety added the couple of extra H-bond donors and the acceptors. Due to this couple of H-bonds donors and acceptors, the TPEG structure shows maximum number of H-bonds. The CPEG structure shows the lowest H-bond forming with solvents. However, the CPEG structure shows the highest no. of H-bonds but it prepares with 5 chains. So, the average number of H-bonds reduces. The CPEG structure self–assembles its own structure in such a way that it blocks its active H-bond sites. Due to this reason, the CPEG structure forms a smaller number of the H-bonds. Comparing the Fig. 4(a-b) the H-bonds between the polymer to the solvents are higher compares to the polymer to polymer. The PEG molecules join with an ether moiety, which is an H-bond acceptor.

3.3. Dynamic swelling ratio

The dynamic swelling ratio of the different hydrogel frameworks is given in Fig. 5. The swelling ratio is arranged in the following order: CPEG > PEG > NPEG > TPEG, given in Fig. 5(a). The swelling ratio of each hydrogel is constant throughout the dynamics. This signifies the hydrogel maintains its structural stability throughout the dynamics, which is suggested by the H-bonding results. The final frame swelling ratio is observed in Fig. 5(b). The swelling ratio of CPEG at the final frame is 446.35, with a standard deviation of 3.37. This is because the CPEG framework attracts both non-freezing and freezing water, as supported by the H-bond results. The network structure of CPEG positions the TPEG chain at the center, with NPEG surrounding and cross-linked, leaving unsaturated hydrocarbons exposed. These unsaturated hydrocarbons are responsible for attracting freezing water. In contrast, the TPEG hydrogel shows the lowest swelling ratio of 172.04, with a standard deviation of 2.48, due to its self-assembly reducing the number of H-bonds. The PEG structure forms the highest fraction of H-bonds with water, attracting more non-freezing water but less freezing water. The NPEG hydrogel has the second-highest swelling ratio of 203.81, with a standard deviation of 8.52. Though NPEG and TPEG attract similar amounts of non-freezing water, the NPEG structure attracts more freezing water due to the presence of unsaturated hydrocarbons, which are more electronegative than saturated ones.

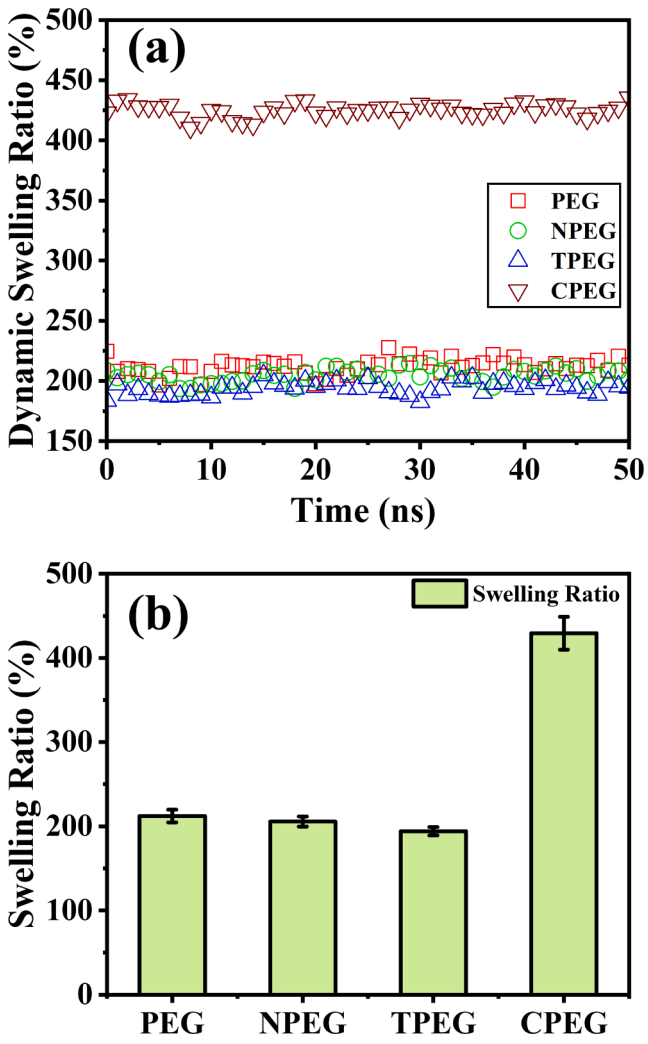


Fig. 5. The dynamic swelling ratio of hydrogels (a) swelling ratio at different time frames (b) swelling ratio of the last frame.

Table 1
Average number of H-bonds.

Polymer name	Average no. of H-bonds between polymer	Average no. of H-bonds between solvents	Percent of H-bonds forming in polymer chain to the polymer chain	Percent of H-bonding forming in polymer chain to solvents
PEG	0.66	93.74	0.70	99.30
NPEG	3.26	84.10	3.73	96.27
TPEG	4.59	84.14	5.18	94.82
CPEG	17.97	302.21	5.61	94.39

3.4. Analysis of RDF

Radial distribution functions investigate the spatial distribution of water inside a framework of polymers as well as the distribution of atoms in the framework of polymers. RDF values generally converge to unity at large separations in a uniform system. However, the existence of solute clusters can interfere with this property. The RDF curves ($g_{O-HW}(r)$) between the oxygen atoms in the polymer backbone and the hydrogen atoms in the water molecule are shown in Fig. 6.

The findings suggest that the existence of H-bonding between PEG

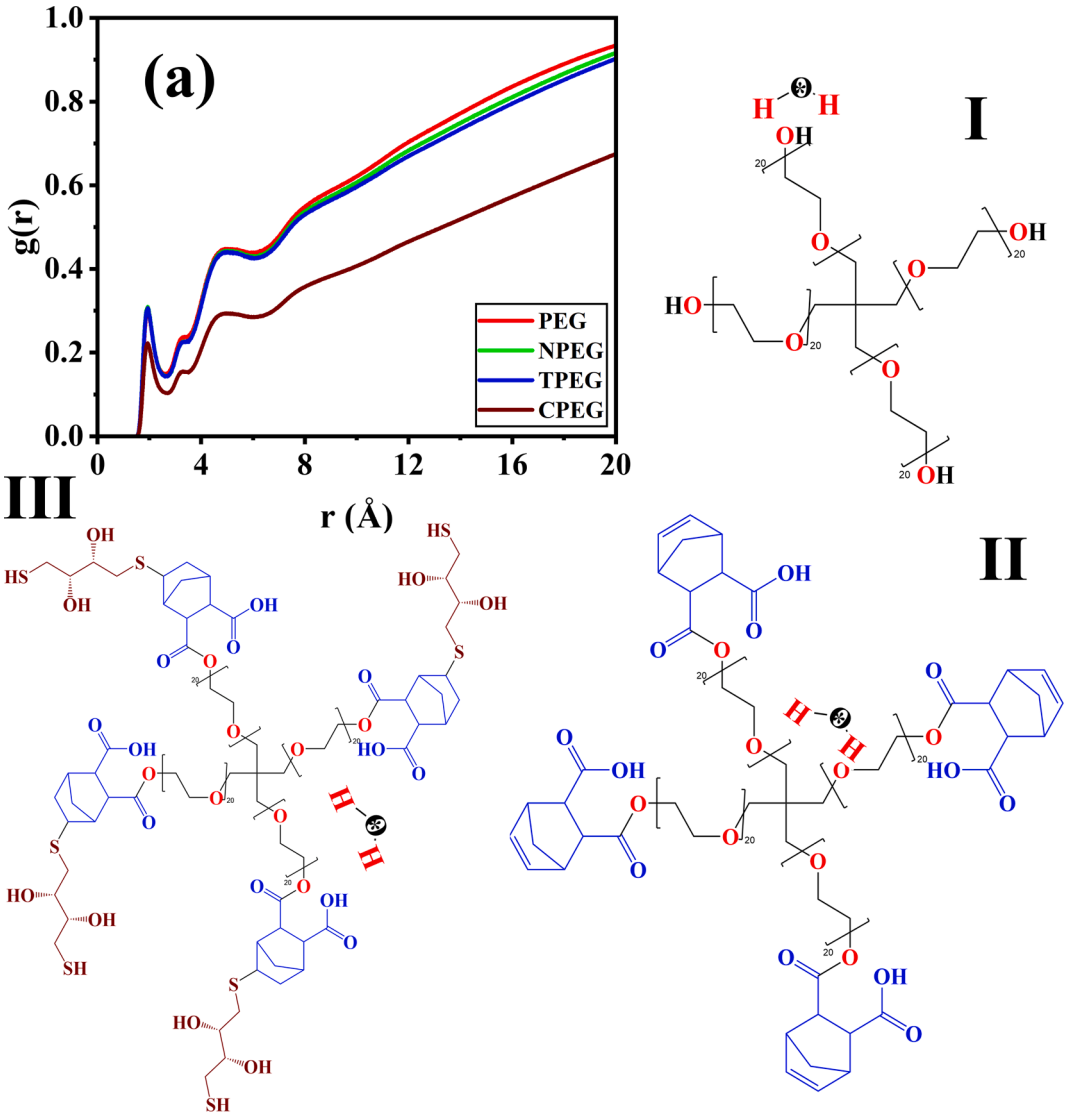


Fig. 6. The RDF of oxygen atoms in the polymer backbone to water hydrogen atoms. (a) RDF of oxygen atoms in the polymer backbone to water hydrogen atoms for all systems I. schematic representation of 4–arm PEG interacting with water, II. schematic representation of 4–arm NPEG interacting with water, III. schematic representation of 4–arm TPEG interacting with water, and IV. schematic representation of 4–arm CPEG interacting with water.

polymer and water is pivotal, influencing the spatial arrangement of particles. The first solvation shell occurs at 0.265 nm, which suggests the H-bonding regime. The order of the RDF is PEG > NPEG > TPEG > CPEG. The value of the RDF is 0.34, 0.32, 0.29, and 0.23, respectively. The value of the coordination number is 0.19, 0.18, 0.16, and 0.12, respectively. The higher RDF peak for PEG implies greater H-bonding between this polymer and water molecules, leading to an increased likelihood of particle presence at specific distances. Correspondingly, higher coordination numbers are observed for PEG compared to other polymers. The H–bonding results also suggest similar phenomena where the fraction of H-bonding with PEG and water is maximum. However, the RDF shows an increasing trend at larger distances, suggesting the presence of water hydrogen atoms in regions with lower density than the system's average bulk density. This phenomenon may arise from spatial correlations between particles, resulting in a lower density at certain distances. The RDF values exhibit sub–unity values at long distances for all systems, indicating the formation of chain clusters. The experimental data show that the PEG hydrogels tend to form aggregated micro-domains[70,71]. The fluorescence studies show that the PEG form aggregated in the microdomain. During fluorescence studies, a PEG fatty acids mixture mixed in a pyridine and a relatively low ratio peak is observed. It is important to emphasize that the $g_{O-HW}(r)$ value remains below one, even at a large distance. Overall, the findings emphasize the crucial role of H-bond formation between PEG and water in determining spatial particle distribution, and RDF behavior reflects interparticle interactions and particle structural properties.

3.5. Mean square displacements (MSD)

The MSD curve of oxygen atoms within the polymer backbone (highlighted in red color) of PEG, NPEG, TPEG, and CPEG is shown in Fig. 7. The MSD value of oxygen atoms in the PEG polymer backbone is higher than that of other polymer chains. It indicates that the oxygen atoms of the PEG backbone experience more significant displacement over time with respect to other oxygen moieties present in the polymer backbone, indicating a more dynamic and flexible behavior. This higher mobility is due to interactions between the PEG chain and water moieties. This is attributed to reduced intermolecular interactions, weaker chain entanglements, and enhanced solvation effects. The H–bond results suggest that the fraction of the H–bond with water is maximum. It

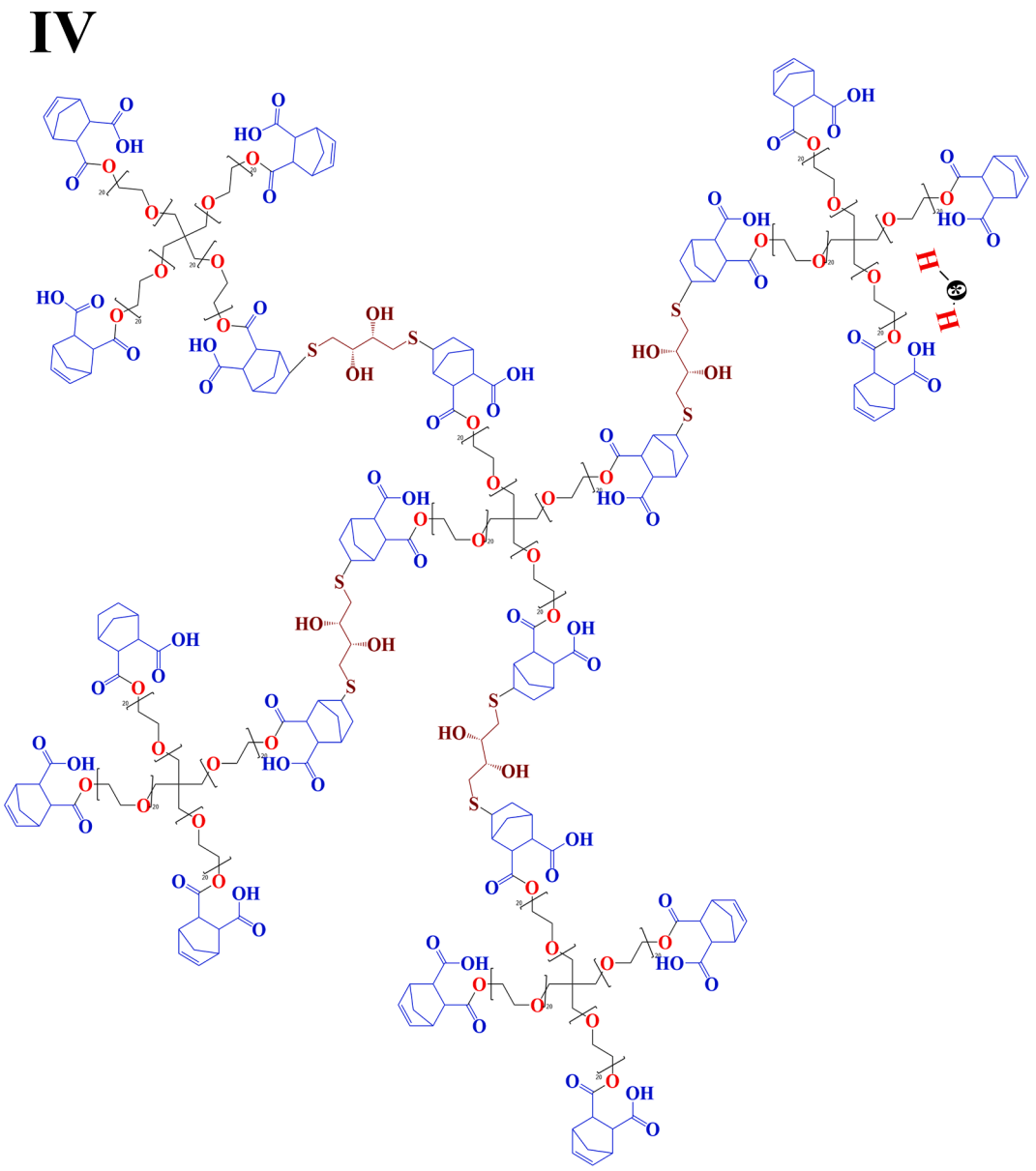


Fig. 6. (*continued*).

signifies that the PEG chain moves freely in the aqueous medium. Fig. 7 (a) depicts linear behavior throughout the dynamics. The logarithmic slope of MSD curve is one which signifies the diffusive behavior of PEG chain in the aqueous media. The self-diffusion coefficients of the PEG chain are 1.97×10^{-10} m^2s^{-1} and the experimental self-diffusion coefficients value is 0.202×10^{-10} m^2s^{-1} [72]. This higher value is observed due to the less water contains in the model. The experimental self-diffusion coefficients of PEG 400 are measured using 3D-nuclear magnetic resonance. The Stejskal-Tanner Equation is used to calculate the self-diffusion coefficients from 3D-nuclear magnetic resonance results. However, the Stejskal-Tanner formula to calculate the self-diffusion coefficients and Einstein's self-diffusion coefficients calculation formulae are different, which may be making it 10 times larger compared to the experimental data. Fig. 7(b) suggest the MSD curve of the oxygen–containing moiety in the NPEG chain. The MSD curve shows linear behavior initially till 10 ns. During this time periods, the logarithmic slope value is less than one which signifies the subdiffusive behavior. As norbornene functionalities with the PEG chain, the number of H-bond increase. After 10 ns, the MSD curve shows linear behavior throughout the dynamics, but the MSD slope decreases. This occurs due to increased non-bonded weak interactions between the moieties. Fig. 7 (c) suggest the MSD curve of the oxygen–containing moiety in the TPEG chain. The MSD curve shows linear behaviour throughout the dynamics. However, the logarithmic slope value suggests less than one, which signifies the subdiffusive behaviour of moieties. This subdiffusive behaviour is due to H-bonding between the moieties. However, the MSD of TPEG is higher compares to the MSD of the NPEG. The dithiothreitol structure contains hydroxyl groups and thiol groups which can form the H-bonds with water. As a result, the dithiothreitol moiety present in the TPEG structure is more water soluble compared to norbornene functionalised NPEG chain. Fig. 7(d) shows the MSD of oxygen–containing moiety in the CPEG chain. Initially, the MSD curve shows linear behavior till 20 ns. During this time period, the logarithmic slope value is less than one which signifies the subdiffusive behavior. The chemically crosslinked 4-arm polyethylene glycol network creates a complex environment that restricts the movement of ether moieties presents in the macromolecules. This confinement effect, coupled with the presence of obstacles and interactions between the polymer chains, leads to anomalous diffusion. After that the MSD curve shows linear behavior till end of the dynamics. During this time periods, the slope of MSD further

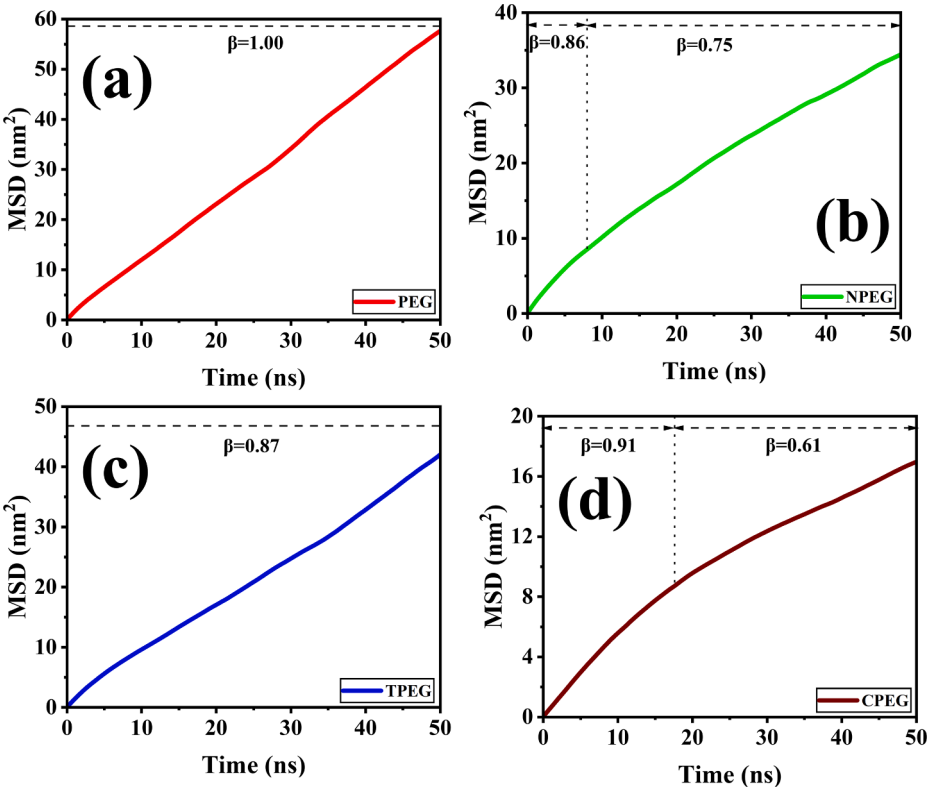


Fig. 7. The MSD curve of oxygen atoms in the polymer backbone. (a) MSD curve of 4–arm PEG, (b) MSD curve of 4–arm NPEG, (c) MSD curve of 4–arm TPEG polymer chain, and (d) MSD curve of 4–arm CPEG polymer chain, I. Schematic diagram of 4–arm TPEG, II. Schematic diagram of 4–arm NPEG, III. Schematic representation of 4–arm PEG and IV. Schematic representation of 4–arm CPEG.

decreases. This occurs due to increased non-bonded weak interactions between the moieties. Comparing Fig. 7(a-d), the NPEG, TPEG, and CPEG systems show subdiffusive behavior throughout the dynamics. As the norbornene and dithiothreitol functanalises with PEG, the number of H-bonds increase between the polymer to polymer. As a result, the agglomerate NPEG, TPEG, and CPEG chain forms temporary physical crosslinks, which restrict the motions.

Fig. 8(a) and 8(b) show the MSD curves for unreacted ene groups in the NPEG chain in the NPEG and CPEG systems, respectively. Both curves exhibit initial linear behavior with log-scale slopes below one, indicating subdiffusive movement due to H-bonding and complex environments. For NPEG, this linear phase lasts until 15.2 ns, while for CPEG it extends to 18 ns. After these periods, both curves remain linear but with decreased slopes. This change stems from increased non-bonded weak interactions between molecules. In CPEG chains, the chemically crosslinked 4-arm polyethylene glycol network restricts ether group movement in macromolecules. This confinement effect, combined with obstacles and polymer chain interactions, results in unusual diffusion patterns. The ongoing slope reduction in both cases reflects the growing influence of non-bonded weak interactions between molecular components as the simulations progress.

3.6. Stress-strain analysis

The stress–strain analysis of the four polymer systems is shown in Fig. 9. The higher yield stress observed in the PEG polymer compared to the other matrices indicates its superior resistance to plastic deformation under applied stress. This suggests that PEG maintains its elastic behavior over a wider range of stress, potentially due to its simpler, more uniform structure allowing for better distribution of stress throughout the material. The TPEG structure, with its higher yield strain compared to the other matrices, demonstrates a greater ability to undergo elastic deformation before transitioning to plastic behavior. This property implies that TPEG can absorb more energy and withstand

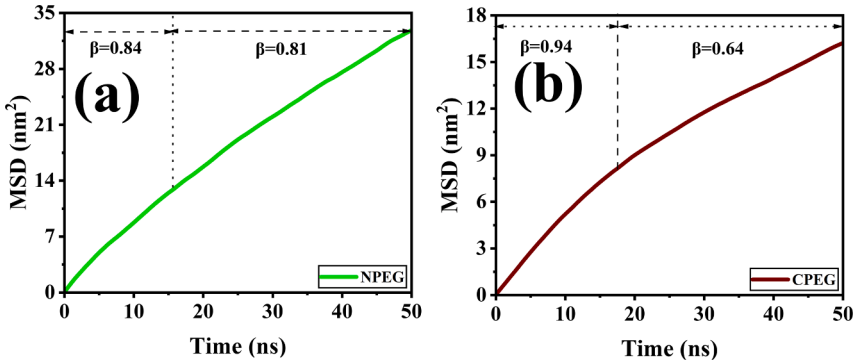


Fig. 8. The MSD curve of unreacted ene groups in the NPEG chain (a) NPEG (b) CPEG.

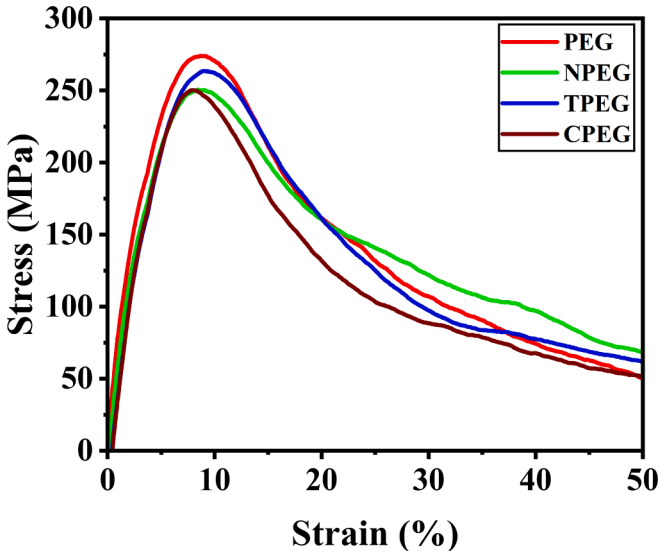


Fig. 9. The stress–strain curve for the all-polymer system.

larger deformations while still returning to its original shape, which could be beneficial in applications requiring flexibility and resilience. The observed order of Young's modulus (NPEG < PEG < TPEG < CPEG) is provided in Table 2. The NPEG macromolecules show lower Young's modulus, suggesting it is the most flexible and easily deformed under stress, likely due to the norbornene functionalization introducing some structural irregularity. PEG's higher modulus compared to NPEG indicates that the basic four-arm structure provides a balance of flexibility and stiffness. TPEG's further increased modulus can be attributed to the additional dithiothreitol-norbornene functionalization, which may introduce more crosslinking points or interactions between polymer chains, enhancing overall rigidity. Finally, CPEG's highest Young's modulus signifies it is the stiffest and most resistant to deformation, which is consistent with its thiol-norbornene crosslinked structure providing the most extensive network of covalent bonds among the four systems. This progression in Young's modulus values reflects the increasing complexity and interconnectedness of the polymer structures, from the relatively simple PEG to the highly crosslinked CPEG. These differences in mechanical properties highlight how structural modifications and crosslinking strategies can significantly influence the behavior of polymer materials, allowing for tailored designs to meet specific application requirements in fields such as biomedical engineering, materials science, and advanced manufacturing.

4. Conclusion

The dynamic behavior of thiol–ene crosslinked 4–arm PEG hydrogels in aqueous media is investigated using all-atom MD simulations. Norbornene significantly stabilizes the 4–arm PEG chains, enhancing hydrogel stability, as demonstrated by RMSD results. RDF analysis reveals the formation of small clusters in the aqueous medium and shows that the first solvation shell of PEG chains with water molecules lies in the H-bonding region, emphasizing the importance of H-bonding in hydrogel properties. Among the polymers, 4–arm PEG exhibits the strongest cohesive properties, while 4–arm TPEG has the highest adhesive properties. Swelling ratio calculations indicate that the 4–arm CPEG structure has the highest swelling ratio, followed by NPEG, which attracts more bound water due to H-bonding. MSD analysis shows that the 4-arm PEG chain exhibits diffusive behavior, with a diffusion coefficient of 1.97×10^{-10} m^2/s, while the other matrices show subdiffusive behavior typical of viscoelastic materials. Stress–strain analysis reveals that CPEG has the highest Young's modulus, indicating its stiffness, and TPEG has the highest yield strain, showing its ability to undergo elastic deformation. These findings offer valuable insights into the structure–property relationships in these hydrogels, enabling better designs for applications such as drug delivery, tissue engineering, and smart materials.

Table 2
The young's modulus data.

System's name	Young modulus (MPa)
PEG	2508.99
NPEG	2213.44
TPEG	2821.54
CPEG	3054.52

CRediT authorship contribution statement

Rontu Das: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Debashis Kundu:** Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Debashis Kundu reports financial support was provided by Science and Engineering Research Board (SERB), India. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Detailed information on simulation systems, initial geometry of polymers is given, dynamic behavior of hydrogels at the different time interval. Supplementary data to this article can be found online at https://doi.org/10.1016/j.commatsci.2024.113516.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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